Know When to Fold: Risk-Controlled Abstention for Protein–Ligand Co-Folding under Novelty Shift

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Abstract

Protein–ligand co-folding models (AlphaFold3, Boltz, Chai) report confidence scores that correlate with pose accuracy, yet a correlation is not a decision rule and it is weakest on the novel pockets and chemotypes that drug discovery cares about. We build `foldgate`, a training-free layer that turns frozen co-folding confidence into a risk-controlled accept/abstain gate with a finite-sample selective-risk guarantee, and we show the guarantee collapses under training-similarity shift: a gate that meets the target error rate on average (marginally compliant) still under-controls error about twofold on novel strata, and calibrating on familiar targets while deploying on novel ones drives realized error to 0.55 (AF3, and 0.46 to 0.61 across the five models) against the 0.20 target, with the guarantee holding in 0 of 300 resampled runs for every one of them. Reliability drift, the movement of $P(\text{correct} \mid \text{confidence})$ along a novelty axis, points to concept shift as the dominant, model-relative mechanism, the confidence-to-correctness map drifting by up to $+0.63$ (Protenix) on structural novelty. We prove why: at fixed coverage the realized target risk obeys an exact identity, a covariate-corrected reference plus an accept-region concept gap that no training-free reweighting can move, so weighted conformal silently violates its own certificate by exactly that gap, and only in-stratum recalibration attains the floor, at the in-stratum-label cost we price. A per-stratum feasibility map sharpens the impossibility, the frozen score's usable coverage decaying to zero on the most novel strata where abstention is the only certified action. We repair what is repairable with group-conditional calibration and a label-free worst-subpopulation (CVaR/DRO) certificate under multiplicity control, and match the certifier to the loss: an exact binomial on the binary pose label, and on the graded RMSD loss a variance-adaptive betting bound that certifies a usable operating point (43% coverage at 1 Å) where a worst-case bound certifies none. Downstream, on a released co-folded screen the governed pose signal enriches actives above docking on post-2022 novel targets (EF@1% 9.3 vs 2.0), and as an abstention gate over a Boltz-2 affinity ranker it preserves enrichment random abstention destroys on 75% of them; on Runs N' Poses the accepted poses recover 0.14 to 0.21 more crystal contacts than the rejected ones. The layer is training-free and runs on released predictions with no GPU, the one cross-dataset check aside.

1 Introduction

Co-folding models predict a protein–ligand complex and emit confidence scores (interface predicted TM-score, ipTM; ranking score; predicted aligned error, PAE). A practitioner who wants to act on a predicted pose needs a decision rule that carries an error guarantee. Conformal selective prediction supplies one: accept a pose when its score clears a threshold, abstain otherwise, and certify that the error rate among accepted poses stays below a target α with high probability [7, 20]. The catch is that this guarantee assumes exchangeability (that a new target is statistically interchangeable with the calibration examples), which breaks precisely where deployment matters, on ligands and pockets dissimilar to the training set. In drug-discovery terms the model is most overconfident on exactly the never-before-seen molecules a program cares about, so the operational question is not the average accuracy but whether a single delivered pose can be trusted.

We make five contributions, all training-free with respect to the co-folding model and all computed from released predictions on CPU. (1) We give a finite-sample selective-risk gate over frozen co-folding confidence and show its guarantee collapses under novelty shift, then measure *reliability drift* to attribute the collapse to concept shift as the dominant (model-relative) mechanism (Sec. 3). (2) We prove an impossibility and a matching achievability result characterizing what a training-free reweighting of a frozen score can control under concept shift, validated against a known conditional and on a public non-co-folding benchmark, and we map its empirical shadow: a per-stratum feasibility frontier that decays to zero on novel strata, strictly stronger than the coverage-pinned theorem (Sec. 4). (3) We repair coverage with group-conditional calibration and a label-free worst-subpopulation CVaR/DRO certificate, with explicit multiplicity control across the five models, and report the target-label budget the repair costs (Sec. 5). (4) We match the certifier to the loss: an exact binomial on the binary pose label, a variance-adaptive betting bound on the graded RMSD loss, which certifies a usable operating point where a distribution-free bound certifies none (Sec. 5). (5) We produce the downstream decision on the pose-confidence signal the gate is calibrated on: on a released co-folded screen, abstaining on unreliable poses (the layer's gate) preserves the enrichment that a Boltz-2 affinity ranker delivers on novel targets, and, on Runs N' Poses, the accepted poses recover more of the crystal binding contacts (Sec. 6).

The closest prior work is CoDrug [14], which conformalizes a *scalar* molecular property under covariate shift with the same weighted-conformal tool [3] our theorem shows concept shift defeats in this domain; we are not aware of a prior conformal treatment of co-folding pose correctness, or of training-set similarity used as the shift variable for it. Confidence-tracks- accuracy studies for co-folding virtual screening [10] rank by ipTM but supply neither a coverage guarantee nor an abstention rule, which is the gap this layer fills. A reweighting-invariant selective-risk floor of the same abstract character was shown independently and concurrently for LLM abstention [18]; our result is its co-folding specialization, localized per novelty stratum with a matching in-stratum achievability bracket. Conformal selection [15, 16] is complementary rather than competing: it certifies which *compounds* to advance given a labeled calibration set, while this layer certifies the *pose* correctness a co-folding screen rests on, training-free, its own screening use being a heuristic transfer because the released decoys carry no pose-RMSD label.

2 The gate

We consider a delivered top-1 pose per system with a native confidence s and a binary label $Y = 1$ iff the symmetry-corrected ligand-RMSD is at most 2 Å. The gate accepts when $s \geq \tau$. We pick the largest accept set (smallest τ) whose selective risk is certified below α by Learn-then-Test with fixed-sequence testing [1], testing each $H_0(\tau): P(\text{error} \mid s \geq \tau) \geq \alpha$ with an exact binomial tail along a pre-specified coverage grid; this gives $P(\text{selective risk} \leq \alpha) \geq 1 - \delta$, finite-sample and distribution-free. We optionally replace s with a calibration-only combined score, a shallow gradient-boosted map from cheap per-prediction features (confidence, interface ipTM, ensemble spread, cross-model agreement, ligand descriptors) to $P(\text{correct})$; the co-folding model is never retrained, and novelty is excluded from the score and enters only in calibration. We evaluate five models (AF3 [11], Boltz-1, Boltz-1x [19], Chai-1 [12], Protenix [13]) on Runs N' Poses [8], 12,602 delivered poses over 2,425 systems, at $\alpha = 0.20$, $\delta = 0.10$ (a released Boltz-2 affinity head, used only as an ungoverned screening comparator in Sec. 6, brings the raw table to 13,535). The selective-risk experiments score one delivered pose per prediction; where a certificate must be per deployed model, we reduce to one pose per (system, model), 11,254 independent target-labels across the five governed models, since several ligand instances of one system fail together and pooling them as independent draws would be anti-conservative.

3 The exchangeability break and its mechanism

The break. We stratify targets by training-set structural and chemical similarity into S_0 (familiar) through S_4 (no analog). A single global threshold hides severe per-stratum under-coverage: for AF3 the marginal realized risk of 0.177 masks 0.375 on S_3 and 0.427 on S_4. The operational failure is starker. When we calibrate the gate on familiar targets and deploy it on the novel region, the realized error is 0.55 while the target is 0.20, and the finite-sample guarantee holds in 0 of 300 resampled runs across all models (Fig. 1). Marginal validity is not enough, which is the whole point.

Why it breaks. Weighted conformal would fix this if the shift were pure covariate shift, where the map $P(Y \mid s)$ is stable and only the marginal $P(s)$ moves. We test that assumption directly by measuring *reliability drift* $D(\nu) = \sum_{\text{bins}} w\,[P(Y \mid s, S_0) - P(Y \mid s, S_\nu)]$, the confidence-conditional

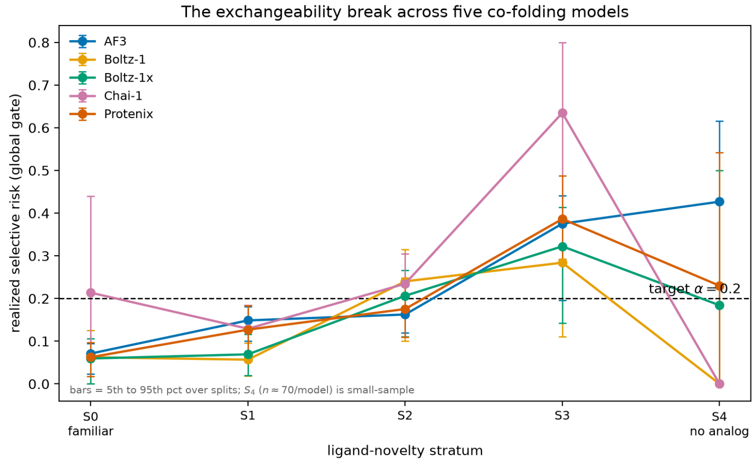

Figure 1: The exchangeability break (the money result). A single global selective-risk threshold, calibrated to look marginally compliant, under-controls error on the novel strata for every model: realized risk climbs well above the target $\alpha = 0.20$ as training-set similarity falls, and calibrate-on-familiar / deploy-on-novel drives it to 0.55 against the 0.20 target with the finite-sample guarantee holding in none of the resampled runs. The break is carried by S_3; the no-analog stratum S_4 ($n \approx 68$ to 76 per model) is small-sample, so a dip there for some models is estimation noise rather than restored safety.

correctness gap between the familiar reference and stratum v. Drift is large and its 90% bootstrap interval excludes zero for every model on structural novelty (Protenix reaches $+0.63$ on pocket novelty, Chai $+0.52$); on temporal novelty it is an order of magnitude smaller ($|D| \leq 0.07$, with only 2 of 15 temporal cells statistically resolved). Confidence is optimistic on novel structure, the signature of concept shift. A signed accept-region decomposition makes this quantitative: for AF3 on S_3 the target risk gap of 0.328 splits into a concept term of 0.309 (90% CI $[0.27, 0.35]$) and a covariate term of 0.018, which the next section turns into a theorem. This split evaluates the source conditional η_P on the target's support, which on genuinely novel pockets is an extrapolation, so the concept-versus-covariate attribution is model-relative rather than identifiable; we report the axis and the base model with it.

4 What a training-free reweighting can and cannot fix

The decomposition above is an instance of a general fact. Fix the frozen score s and the accept rule $\{s \geq \tau\}$. For this section only we take $Y=1$ to mean *error* (the complement of the gate's correctness label), so that risks read directly. Let $\eta_P(x) = P(Y=1 \mid X=x)$ and η_Q be the error conditionals on the calibration law P and the deployment law Q, and call a reweighting *training-free* if its weights are covariate-measurable and use no target labels (density-ratio, classifier, or KLIEP weights, including the point mass of weighted conformal). Write $R_{\text{ref}}(\tau_c) = \mathbb{E}_Q[\eta_P \mid s \geq \tau_c]$ for the covariate-corrected reference at coverage c and $\bar{\Delta}_c = \mathbb{E}_Q[\eta_Q - \eta_P \mid s \geq \tau_c]$ for the accept-region concept gap.

> **Theorem (impossibility, informal; App. A).** At fixed coverage c the realized target selective risk is $R_Q(\tau_c) = R_{\text{ref}}(\tau_c) + \bar{\Delta}_c$, and $\bar{\Delta}_c$ is *invariant* to the choice of training-free weights. Hence (i) the oracle-weighted plug-in certificate reports R_{ref} and still under-states the realized risk by $\bar{\Delta}_c$; and (ii) level α is unachievable at coverage c by any training-free reweighting whenever $\bar{\Delta}_c > \alpha - R_{\text{ref}}(\tau_c)$.

In words: the part of the error that comes from confidence meaning something different on novel molecules is exactly the part no unlabeled reweighting can see or remove, so a certificate built without novel labels understates the true risk by precisely that amount.

The proof (App. A) is a tower-property identity: a covariate-measurable weight recovers only the *source* conditional η_P and cannot reach η_Q, which completes the covariate-shift boundary of weighted conformal [3] and specializes the irreducible term of domain-adaptation lower bounds [22]; a reweighting-invariant selective-risk floor of the same character was shown independently for LLM abstention [18]. Matching achievability: in-stratum recalibration attains the floor R_{Q_g} given n_g stratum labels, with slack $O(\sqrt{\log(1/\delta)/n_g})$; pointwise conditional coverage being impossible [6], the guarantee is neighborhood-conditional.

Validation on a general benchmark. A synthetic generator with a known $P(Y \mid s, \nu)$ gives ground truth for each claim. With genuine target drift, $R_{\mathrm{ref}} = 0.273$ and $\bar{\Delta} = 0.205$ sum to the realized $R_Q = 0.478$, the realized risk is invariant across nine reweightings (residual < 0.001), the oracle certificate under-reports by $\bar{\Delta}$, and above the crossing no reweighting reaches α; in-stratum recalibration controls only with target-stratum labels, and an error floor above α forces abstention. A shared-label variant reproduces the decomposition and invariance with a near-zero pooled gap, placing the concept effect at the worst stratum as in the co-folding break. On public benchmarks with no co-folding, a single-threshold gate under-controls the worst novelty block, weighted conformal issues a certificate its realized risk then violates, and group-conditional calibration certifies control by abstaining on the blocks it cannot cover (App. B): on elec2 (worst-block concept gap 0.36) the realized worst-block risks are 0.31, 0.34, and 0.20, and on ACS Income, whose worst-block concept gap is only 0.05 so the shift is covariate-dominated, weighted conformal already repairs the worst state ($0.23 \to 0.20$).

The empirical shadow: a feasibility frontier. The theorem is pinned at a fixed coverage and leaves open reaching α by lowering it, so we measure that directly. We fix the deployed rule on the familiar stratum and, per target stratum, report the largest source coverage $c^\star_g = \max\{c : R_{Q,g}(\tau(c)) \leq \alpha\}$ at which it still holds α, one delivered pose per (system, model). The frontier decays with novelty and then vanishes: for AF3 on the ligand axis $c^\star$ runs 1.00, 0.85, 0.75, 0.20, 0.00 across S_0 to S_4 at $\alpha = 0.20$, and 0.65, 0.40, 0.10, 0.00, 0.00 at $\alpha = 0.10$. Requiring at least 20 accepted targets to call a stratum feasible, 21 of 50 model-by-stratum cells admit no operating point at any coverage at $\alpha = 0.20$ and 26 of 50 at $\alpha = 0.10$; the claim survives an exact Clopper-Pearson lower bound on $R_{Q,g}$ at every measurable coverage in 35 of the 47 pooled zero-frontier cells. A zero frontier is strictly stronger than the coverage-pinned theorem, which speaks only at the deployed coverage, and it is an empirical finding rather than a corollary. The deployed view is starker: the AF3 rule that Learn-then-Test certifies at $\alpha = 0.20$ on familiar targets realizes risk 0.538 on S_3, a margin of -0.338, so the certificate is inverted, not merely loose.

A one-sided escape is not forbidden by the theorem, which constrains only certified *upper* bounds. Cross-model pose disagreement supplies a label-free *lower* bound on ensemble risk by triangle inequality to the crystal pose, extending an oracle-free disagreement floor known for discrete outputs [24, 25] and the older folklore that pose agreement filters wrong poses [26] to a continuous structured target. It does rise with novelty, but it is blind to consensus error, where all models share one wrong mode; the consensus regime covers 64% of the analyzed complexes at risk 0.152 against 0.698 on the diverse remainder (marginal), and the error it hides, the consensus rate times the risk inside it, grows from 0.058 on S_0 to 0.120 on S_3 (AF3, accept region at 50% coverage) even as the consensus mass shrinks. The blind spot was predicted as a corollary before it was measured, and it is exactly the novel-chemotype regime the certificate most needs, so the escape is real and narrow; we develop it and its metric construction separately.

5 Repairing the guarantee

We repair what the theorem allows in three moves: a group-conditional gate that restores per-stratum control given stratum labels, a label-free worst-subpopulation certificate for when the novel group is unnamed, and a certifier matched to the loss so the label cost is minimal; a measured drift diagnostic then says which repair a deployment region admits.

Group-conditional calibration [17] keys a separate threshold per novelty stratum, restoring per-stratum control at an honest coverage cost (with the combined score AF3 accepts 52% of S_1 and 39% of S_2 at risk $\leq \alpha$, against 8% and 12% natively; Fig. 3). It needs stratum labels, which Runs N'

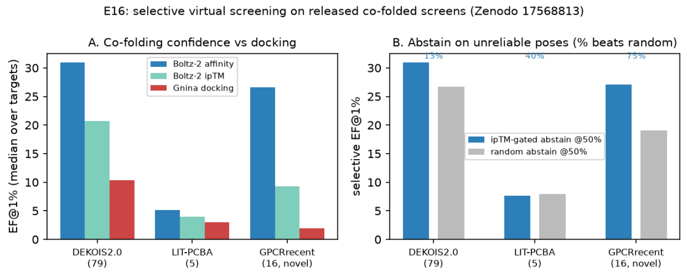

Figure 2: Selective virtual screening on released co-folded screens. (A) The pose signal (ipTM) the layer governs enriches actives above docking, with the ungoverned Boltz-2 affinity head shown for comparison; the gap is largest on the post-2022 novel GPCR targets. (B) Ranking by the affinity head and abstaining on low-ipTM poses at 50% coverage preserves enrichment that random abstention destroys; percentages are the fraction of targets whose selective EF beats the random-abstention 95th percentile.

Table 1: Selected results. AURC (area under the risk-coverage curve, lower is better) is out-of-fold under target-grouped (leakage-free) cross-validation; LOTO is leave-one-target-out; positive drift D means native confidence is optimistic on the novel stratum; $m^\star$ is the smallest accepted-mass fraction certified at α by the label-free CVaR bound (smaller is better). †Chai native ranking certifies almost no coverage alone ($< 1\%$, so the .30 is at negligible coverage, not a violated guarantee), an honest limit the combined score recovers.

	AF3	Boltz-1	Boltz-1x	Chai-1	Protenix
AURC native $\rightarrow$ combined	.185$\rightarrow$.118	.234$\rightarrow$.162	.215$\rightarrow$.154	.247$\rightarrow$.146	.281$\rightarrow$.166
LOTO realized risk ($\leq$.20)	.185	.157	.168	.30†	.191
reliability drift D, pocket S_3	+.47	+.47	+.47	+.52	+.63
coverage to certify $m^\star \leq .5$	.45	.20	.25	.40	.25

Poses ships. Where the marginal gate over-shoots α on the novel strata, the per-stratum gate holds risk at or below it and, on the no-analog tail where no certified slice remains, folds to zero coverage rather than issue a false certificate.

A label-free worst-subpopulation certificate. When the novel subpopulation is unnamed, we control the conditional value-at-risk of the accepted loss, the mean error of its worst β-fraction. For the binary label, $\mathrm{CVaR}_\beta = \min(1, r/(1-\beta))$ with accepted error rate r, so a finite-sample upper bound on r [2] yields a CVaR_β bound at every β, and those with $\beta \leq \beta^\star$ certify $\mathrm{CVaR}_\beta \leq \alpha$ [4]. By CVaR–DRO duality this certifies that every accepted subpopulation of mass at least $m^\star = 1 - \beta^\star$ has error $\leq \alpha$, with no stratum labels and no importance weights. The combined score certifies $m^\star \leq 0.5$ (every half-subpopulation) at 20–45% coverage where the native score often certifies nothing, and is largely vacuous on deploy-to-novel ($m^\star \rightarrow 1$ for most models), which we report. The same certificate reparameterizes to a chi-square robustness radius $\rho^\star$ [21], and a Bonferroni δ/K correction yields one certificate jointly across all five models (at 20% coverage, $m^\star \leq 0.52$ and $\rho^\star \geq 0.10$, the worst case over models, at $\delta/K = 0.02$ each).

Multiplicity across models. Joint certificate claims (the gate, leave-one-target-out validity, the CVaR/DRO bounds) take a Bonferroni union bound at $\delta/K = 0.02$; the joint leave-one-target-out gate then holds for AF3, Boltz-1, and Boltz-1x with real coverage while Chai and Protenix satisfy it only by abstaining. "For every model" significance conjunctions are intersection-union tests, valid without penalty. The 55-cell drift grid (5 models $\times$ 11 axis-strata) uses a Romano-Wolf step-down bootstrap [28] at family-wise 0.10; 42 cells survive, all 40 structural and 2 of 15 temporal (adjusted $p = 0.053$, both failing the stricter simultaneous band), while no temporal cell passes the 0.05-margin equivalence test, so only the temporal-versus-structural magnitude contrast is robustly supported.

Match the certifier to the loss. Group-conditional repair spends target labels, and how many depends on what is being certified. For the binary pose label the loss is Bernoulli, its count is a

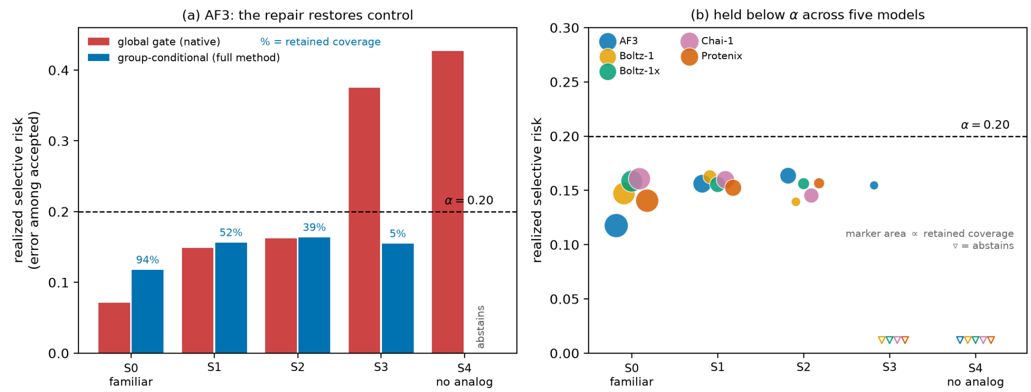

Figure 3: The repair. Group-conditional calibration holds the realized selective risk at or below the target $\alpha = 0.20$ on the strata where the marginal (global) gate over-shoots. (a) AF3: the global gate (native score) exceeds α on the novel strata S_3 (0.38) and S_4 (0.43), while the full method (combined score, group-conditional) holds risk $\leq \alpha$ across S_0–S_3 at the annotated retained coverage and folds entirely on the no-analog stratum S_4. The coverage cost is real and grows with novelty (94% of familiar poses retained, 5% on S_3). Risk control ($\leq \alpha$) rests on the group-conditional calibration; the combined score is what turns the native gate's abstention into that retained coverage (native-score group-conditional folds on S_3). (b) The repaired gate across all five models; marker area is the retained coverage and $\triangledown$ marks strata where the gate abstains. Realized risk stays $\leq \alpha$ wherever a certified slice remains.

sufficient statistic, and the exact binomial tail is that statistic's exact inversion, so no concentration bound improves on it: on the feasible strata it certifies at a median of 38 independent target-labels against 60 for a Hoeffding-Bentkus bound and 102 for pure Hoeffding, and it is the only one of the four that certifies every feasible cell. No variance-adaptive bound can beat it here, since the exact binomial is already the exact inversion of that sufficient statistic, though a variance-adaptive betting bound does improve on the worst-case Hoeffding bound (cheaper in 12/12 cells, median 49 against 102 labels) and is anytime-valid. Variance adaptivity is essential on the *graded* loss. Replacing the 2 Å cutoff with the capped RMSD $\min(\mathrm{RMSD}, 4)/4$ and certifying a mean-RMSD target with a betting bound [27], AF3 certifies 43% coverage at a 1 Å target where the worst-case (Hoeffding) bound certifies essentially none (0.04%), with realized capped-mean RMSD 0.90 Å and empirical guarantee coverage 0.90; the other models certify 2–25% where Hoeffding certifies zero. The reliability layer is therefore not tied to the 2 Å convention, and the certifier follows from the sufficiency structure of the loss rather than from a default.

Reliability drift selects the repair. Reliability drift $D(\nu)$ (Sec. 3) is a label-based diagnostic that tracks the accept-region concept gap $\bar{\Delta}_c$ the impossibility names, so it decides, from a labeled probe rather than after a failed deployment, which repair a target region admits. The theorem says a training-free weighted gate under-reports its realized risk by exactly $\bar{\Delta}_c$: a small drift signals a covariate-dominated shift that weighted conformal repairs, a large drift signals a concept-dominated one where weighted conformal silently violates its own certificate, leaving group-conditional recalibration as the only valid repair. The measured values separate the regimes. On the moderate ligand shift $S_0, S_1 \to S_2$, where the drift is small, the weighted plug-in pulls realized error to the target with no target labels (AF3 0.27 $\to$ 0.19 at 70% coverage), and the temporal axis, whose drift is an order of magnitude smaller ($|D| \leq 0.07$), produces no break to repair. On structural novelty the drift reaches $+0.47$ to $+0.63$, and deploying onto the novel tail $\to S_3, S_4$ the weighted plug-in cannot reach α (naive 0.37–0.50, weighted 0.27–0.38 across the five models), exactly where group-conditional holds. Drift is a heuristic selector and not a guarantee: it needs a labeled probe to measure, and a mis-specified stratifier leaves residual within-stratum drift, so we use it to choose the repair, not to certify one.

Weighted conformal serves as a label-free complement that abstains on real novel data (Sec. 4), so group-conditional calibration stays the operative guarantee. A training-free surrogate that keys the gate on the model's own ensemble disagreement recovers control on moderate novelty but stays a weak proxy on the extreme.

6 Utility and a downstream decision

Ranking utility, validated without leakage. On a held-out test split, at the $\alpha = 0.20$ guarantee the combined score keeps 71% of AF3 poses, 82% of them binding-mode correct, where a native-confidence gate at the same certified error keeps only 22%; for Chai and Protenix, whose native ranking certifies almost nothing (1% and 4% coverage), the same gate recovers 60% and 51%. This survives a strict leakage audit. A pooled split across models would leak 95.1% of test targets into calibration, and even a per-model random split duplicates the roughly 10–11% of systems with several ligand instances, so we re-evaluate out-of-fold under target-grouped (leakage-free) cross-validation: the risk-coverage area still falls 28–41% (AF3 $0.185 \to 0.118$, Protenix $0.281 \to 0.166$; Table 1), the leakage-free leave-one-target-out gate accepts 52% of AF3 poses at realized risk $0.185 \leq \alpha$ (5/5 folds), and the advantage survives a target-cluster bootstrap. The combined score is an optional calibration-only map, not a retrained model and not a separate contribution: the finite-sample guarantee and the impossibility theorem (Sec. 4) hold on the native confidence alone, and the combiner only improves the operating point.

The gate improves a non-circular quality, not just the label it certifies. Accepted-pose purity is arithmetically 1 minus the selective risk, so it cannot show the gate buys anything past the guarantee. Interaction-fingerprint recovery, the fraction of the crystal's protein-ligand contacts the pose reproduces, is a different quantity a medicinal chemist reads and is only correlated with the 2 Å label. Under the gate at $\alpha = 0.20$ the accepted poses recover 0.90 to 0.91 of crystal contacts against 0.71 to 0.76 for the rejected ones, a per-model gap of 0.14 to 0.21 whose 90% bootstrap interval excludes zero for every model. Abstention keeps the poses that recover the right interactions, which is the property downstream structure-based work actually needs.

The decision number. We reuse a released co-folded virtual screen [10], which established co-folding ipTM as a screening ranking signal and its degradation on novel targets, for per-compound confidence and active/decoy labels on DEKOIS2.0 (79 targets), LIT-PCBA (5), and GPCRrecent (16 post-2022 novel targets), with no GPU and no crystal coordinates. Our contribution on this screen is the abstention rule layered over that signal, not the base enrichment; Table 2 places the layer's accept/abstain gate head-to-head against the Hou-style ipTM ranking without abstention on identical targets. We report the pose confidence (ipTM) the reliability layer governs and keep the Boltz-2 affinity head as an ungoverned comparator (Fig. 2). The governed pose signal enriches actives above docking, and the gap widens on novelty: EF@1% is 20.7 against Gnina 10.3 on DEKOIS, and 9.3 ($[5.7, 14.8]$) against Gnina 2.0 and Glide 3.8 on the post-2022 GPCR targets, an interval disjoint from docking's $[1.5, 2.6]$; the affinity head reaches 26.6 but carries no guarantee, and AF3's pose ipTM (25.8 on DEKOIS) only widens the pose-over-docking gap. For selective screening we rank by the affinity head and abstain on low-ipTM poses, so the layer supplies the abstention while the affinity head supplies the ranking, preserving enrichment that random abstention destroys on 75% of the novel GPCR targets against 13% on saturated DEKOIS. The layer, calibrated on pose correctness, transfers here as a heuristic abstention rule without a coverage guarantee, since the screen carries no pose-RMSD label. Enrichment degrades on novel actives: the affinity head trends from 27 on the most novel similarity bin to 42 on train-similar ($n \approx 16$, CIs overlap), and the governed pose signal sits lower on novel actives (11) than familiar (18), the screening echo of Sec. 3. A Murcko-scaffold split of the DEKOIS actives shows enrichment lower on novel scaffolds for every method, recomputed rather than inherited; because the released decoys ship no structures we cannot correct for analog bias, so absolute EF may be inflated and the pose-over-docking gap and novelty trend carry the claim.

7 Cross-dataset generality and limitations

A threshold frozen on Runs N' Poses and deployed on FoldBench [9] holds for AF3 ($0.15 \leq \alpha$) and under-controls for Protenix (0.25), an honest transfer limited by the confidence fields FoldBench

Table 2: Head-to-head on the released co-folded screens [10], EF@1% (median, 90% target-bootstrap CI). Rows above the rule are ranking without abstention (the Hou-style baseline); the two indented rows add the ipTM accept/abstain gate at 50% coverage over the affinity ranker and the matched random-abstention control. The pose (ipTM) signal is the one the conformal layer governs; the Boltz-2 affinity head is an ungoverned comparator (‡, no coverage guarantee), shown for context. Bottom row: fraction of targets whose selective EF beats the random-abstention 95th percentile. GPCR is the post-2022 novel set; LIT-PCBA CIs are wide at n=5. On the screen the gate is a heuristic transfer: decoys carry no crystal pose, so the RMSD label it is calibrated on is unavailable and no coverage guarantee is claimed here (the selective win is against random abstention, not the affinity head).

EF@1%	DEKOIS (79)	GPCR novel (16)	LIT-PCBA (5)
Docking (Gnina)	10.3 [10.3, 15.5]	2.0 [1.5, 2.6]	3.0 [0, 7.7]
Pose ipTM, no abstention (Hou)	20.7 [18.1, 20.7]	9.3 [5.7, 14.8]	4.0 [0, 26.1]
Affinity head, no abstention‡	31.0 [28.4, 31.0]	26.6 [19.3, 35.7]	5.1 [4.9, 23.1]
+ ipTM gate, 50% cov. (ours)	31.0 [28.4, 31.0]	27.1 [18.8, 35.7]	7.7 [4.9, 18.7]
+ random abstention (control)	26.7 [25.7, 27.6]	19.0 [14.3, 25.2]	8.0 [2.4, 12.3]
targets: selective > random$_{95}$	13%	**75**%	40%

releases. Regenerating the FoldBench Protenix predictions recovers the interface-ipTM field the release withholds and turns this into a feature-matched cross-dataset positive: on 436 protein-ligand targets (384 train-similar, 52 low-homology, ligand-RMSD self-scored against the deposited assembly so feature and label come from one run), the frozen Runs N' Poses interface-ipTM gate transfers with a risk-coverage AURC of 0.380, ahead of the matched `ranking_score` control at 0.454. That same frozen threshold accepts 5% of FoldBench against 26% at home, the coverage collapse of Sec. 3 reproduced across a second dataset and a different confidence source. This regeneration is a comparison of gates, not a re-benchmark of Protenix: our self-scored top-1 success is 0.40 against FoldBench's released 0.57, a gap attributable to the scoring convention, the model version, and the MSA source, which does not affect the transfer because feature and label are self-consistent within the one run. The break is structural, and the theorem and its repair generalize beyond co-folding to the public elec2 and ACS benchmarks (Sec. 4; details in App. B). The no-analog stratum ($n \approx 63$ per model) is uncertifiable under every method, where the only action is abstention, and a randomly-localized gate keyed continuously on novelty [5] still leaves that tail. The co-folding coverage guarantee rests on one full retrospective dataset (Runs N' Poses, five governed models) and one partial single-model regenerated one (FoldBench Protenix), all retrospective. The impossibility and achievability theorem is dataset-independent, and the empirical arm behind it spans 12,602 poses across five co-folding models on that full benchmark plus the regenerated single-model cross-dataset check. A full second model panel and a prospective screen are the natural next step and would widen the empirical claim further.

8 Conclusion

Co-folding confidence becomes a risk-controlled decision only when the guarantee is audited where it fails. On structural novelty a marginally valid gate under-controls error about twofold and holds in 0 of 300 runs, and we prove why with an exact identity: the concept gap that opens there closes under in-stratum recalibration but under no training-free reweighting. The repair holds risk at the $\alpha = 0.20$ target while keeping 71% of AF3 poses where the native gate keeps 22%, and the governed pose signal enriches actives more than fourfold over docking on post-2022 novel targets (EF@1% 9.3 against 2.0). The break and its repair reproduce across five co-folding models, on the frozen models' released predictions and CPU alone.

Data and code availability. The analysis code and calibration tables are public at `https://github.com/rikhinkavuru/foldgate`. A one-command `make repro` reproduces the Runs N' Poses tabular results; the screening arm, the cross-model floor, and the structure-feature results are opt-in (`-screening`, `-structures`) because they consume the larger released artifacts rather than the tabular bundle. Runs N' Poses is at Zenodo (`10.5281/zenodo.14794785`); the screening scores are from **(author?)** [10]. The one cross-dataset check regenerates a Protenix confidence field the FoldBench release withholds, the sole GPU step; those regenerated predictions and their self-scored labels are archived at Zenodo (`10.5281/zenodo.21417241`).

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A Proof of the impossibility, and the achievability bracket

Setup. Frozen score $s = s(X)$, novelty $\nu = \nu(X)$, error label $Y \in \{0,1\}$, accept region $A_\tau = \{s \geq \tau\} \in \sigma(X)$. A training-free reweighting has weights $w = w(X) \geq 0$ that are $\sigma(X)$-measurable and use no target labels. Assume s has no atom at the operating threshold, so for each coverage c there is a unique τ_c with $Q(s \geq \tau_c) = c$; the real ipTM and ranking scores carry ties, handled by boundary randomization, under which the equalities below become $\leq$ / $\geq$ brackets.

Lemma (reweighting sees only the source labels). For any such w, the reweighted plug-in selective risk is $\widetilde{E}^w(\tau) = \mathbb{E}_P[wY\mathbf{1}_{A_\tau}]/\mathbb{E}_P[w\mathbf{1}_{A_\tau}] = \mathbb{E}_{Q_w}[\eta_P \mid A_\tau]$, with $dQ_w \propto w\,dP_X$. *Proof.* w and $\mathbf{1}_{A_\tau}$ are $\sigma(X)$-measurable, so tower over X: $\mathbb{E}_P[wY\mathbf{1}_{A_\tau}] = \mathbb{E}_P[w\eta_P\mathbf{1}_{A_\tau}]$; normalize. The target conditional η_Q cannot appear. □

Theorem. At coverage c: (a) $R_Q(\tau_c) = R_{\text{ref}}(\tau_c) + \bar{\Delta}_c$ with $R_{\text{ref}}(\tau_c) = \mathbb{E}_Q[\eta_P \mid A_{\tau_c}]$ and $\bar{\Delta}_c = \mathbb{E}_Q[\eta_Q - \eta_P \mid A_{\tau_c}]$; (b) any w realizing coverage c accepts exactly A_{τ_c}, so its realized risk equals $R_Q(\tau_c)$, independent of w; (c) with oracle weights the certificate of the Lemma equals $R_{\text{ref}}(\tau_c)$, so it under-reports the realized risk by exactly $\bar{\Delta}_c$; (d) level α is achievable at coverage c by some training-free reweighting iff $R_Q(\tau_c) \leq \alpha$, i.e. $\bar{\Delta}_c \leq \alpha - R_{\text{ref}}(\tau_c)$. *Proof.* (a) $A_{\tau_c} \in \sigma(X)$, so $R_Q(\tau_c) = \mathbb{E}_Q[\eta_Q \mid A_{\tau_c}]$; add and subtract $\mathbb{E}_Q[\eta_P \mid A_{\tau_c}]$. (b) The no-atom assumption pins τ_c from c; realized risk is a functional of (Q, τ_c) alone. (c) Lemma with $w^\star = dQ_X/dP_X$ gives $Q_{w^\star} = Q$, so the certificate is R_{ref}; subtract (a). (d) is (a)+(b). □

This is coverage-pinned: it leaves open reaching α by lowering coverage, and it binds only the class of scalar thresholds of a fixed score with covariate-measurable weights. Re-scoring on ν or a per-stratum threshold escapes it, which is the achievability side: in-stratum split calibration [20] gives stratum-conditional miscoverage $\leq 1/(n_g{+}1)$ regardless of drift, conformal risk control [23] gives the same slack for the joint accept-and-err rate, and the conditional selective risk (a ratio with a random denominator, to which the clean $1/(n_g{+}1)$ does not apply) is controlled with high probability by Learn-then-Test with slack $O(\sqrt{\log(1/\delta)/n_g})$. Per stratum the impossibility floor and the achievability ceiling meet at $R_{Q_g}(\tau_c)$, the frozen score's own in-stratum accept-region risk, reachable by stratified recalibration and provably not by covariate reweighting. The label-free ceiling of the DRO/CVaR certificate matches the worst case over the assumed ambiguity ball and is the honest fallback when no stratum labels are available, vacuous exactly when the ball fails to cover the true drift, as it does on deploy-to-novel.

B Benchmark and validation details

The synthetic generator draws a stratum k from a tilted ν-marginal, a score logit $z \sim \mathcal{N}(m_0 - E\nu_k, \sigma^2)$, and a label $y \sim \text{Bernoulli}(\pi)$ with $\pi = \varepsilon + (1-2\varepsilon)\,\text{sigmoid}(\beta_0 + \beta_1 z - D_{\text{eff}}\nu)$, where $D_{\text{eff}} = D$ on the calibration law and $D + D_q$ on the target law. The knobs isolate covariate shift (E, tilt), genuine concept drift (D_q), and an irreducible error floor (ε); every theorem quantity is available in closed form by quadrature and cross-checked against a Monte-Carlo draw. The real benchmarks reduce any classifier to the triple (s, y, ν) with s its confidence, $y = \mathbf{1}[\text{correct}]$, and ν the natural shift axis (temporal block for elec2, target state for ACS Income); the base classifier is a gradient-boosted tree, torch-free, and the split is grouped by the exchangeable stratum. Realized worst-stratum selective risk at $\alpha = 0.20$: elec2 marginal 0.31, weighted-conformal 0.34, group-conditional 0.20; ACS Income marginal 0.23, weighted-conformal 0.20, group-conditional 0.20. The reliability drift that separates the two cases is 0.36 (elec2, concept) against 0.05 (ACS, covariate).

A benchmark-integrity note for cross-model pose geometry on Runs N' Poses. Recomputing pose RMSD in a common frame carries a silent trap that anyone comparing predicted poses to the released structures will hit. Runs N' Poses ships only the system's receptor chain while co-folding models predict the full biological assembly, so a chain-ordinal superposition can land the ligand in the wrong protomer of a homodimer: the backbone still fits (the superposition residual stays near 0.3 Å) while the ligand sits tens of Ångström away. Neither the residual nor a triangle-inequality consistency check detects it, since a displaced pose is far from everything and satisfies the inequality vacuously, and a distance-consistency check passed at $0/13{,}146$ while 21% of recomputed distances and 15% of binary correctness labels were wrong. Only comparison against the shipped BiSyRMSD label exposes it. Restricting to single-chain receptors with an unambiguous ligand copy raises agreement from Spearman 0.620 to 0.995. We flag this for reuse; a symmetry-aware chain assignment is the general fix.